



Nonlinear Programming

Prof. Karl Ezra Pilario

EgyE 332

1s AY 2026-2027

Outline

1. Motivational Example
2. Representative Algorithms
 - a. Newton Methods (Unconstrained NLP)
 - b. Evolutionary Algorithms (Unconstrained NLP)
 - c. Lagrange Multipliers (Constrained NLP)
 - d. IPOPT (Constrained NLP)
3. Examples in Energy Engineering
 - a. Nonlinear Economic Dispatch
 - b. Economic Dispatch with 2D Valve-Point Effects
 - c. Wind Farm Layout Optimization

Nonlinear Programming

Solar PV Max Power Output

https://github.com/kspilario/EgyE332_opti/blob/main/NLP%20Problems/solar_max_power.ipynb

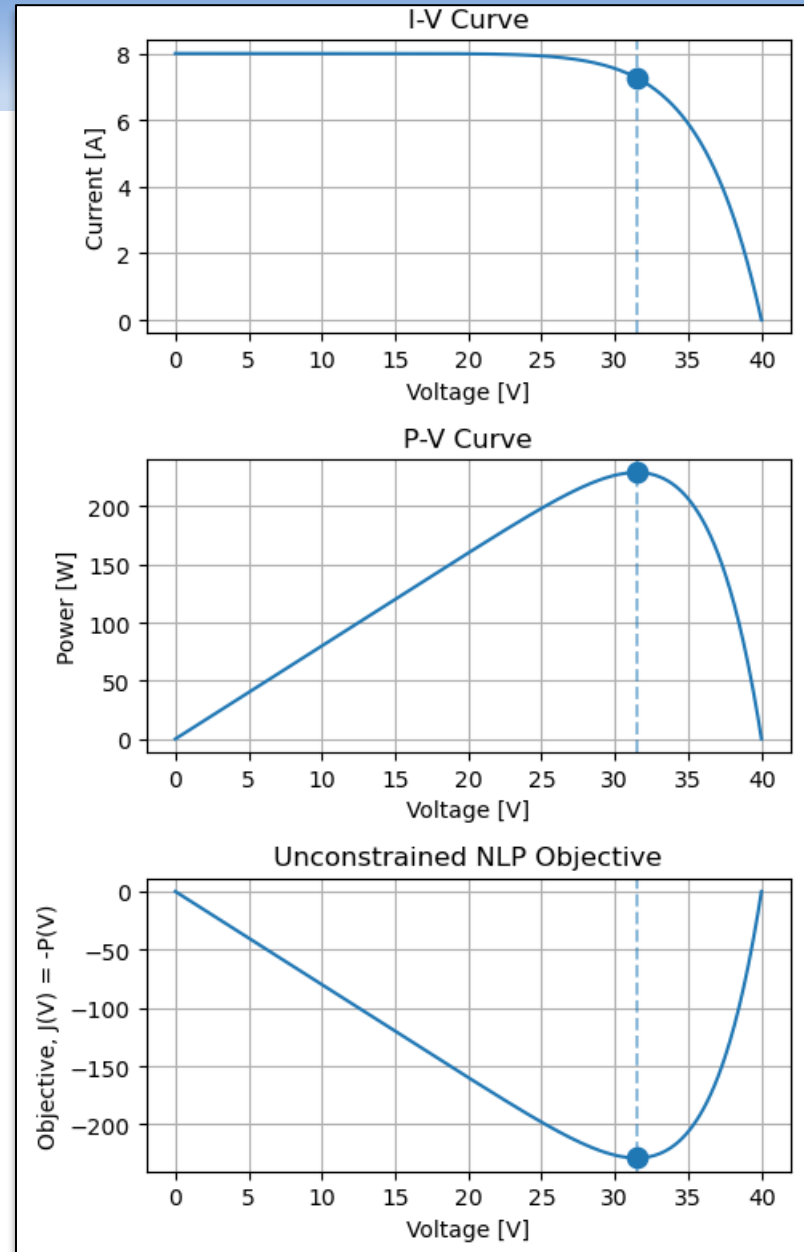
The electrical output of a solar PV depends on the voltage at which it is operated.

For a given irradiance and temperature, a PV module has an I–V characteristic, $I = f(V)$. A simple approximation is given by:

$$I(V) = I_{SC} \left[1 - \left(\frac{V}{V_{OC}} \right)^a \right]$$

Where I_{SC} is the short-circuit current, V_{OC} is the open-circuit voltage, and a is a shape parameter.

Since $P = IV$, we need to solve for the V that gives the maximum power P given a fixed a , I_{SC} and V_{OC} .



Mathematical Programs

In general, the method for solving mathematical models is highly dependent on the **type of model**:

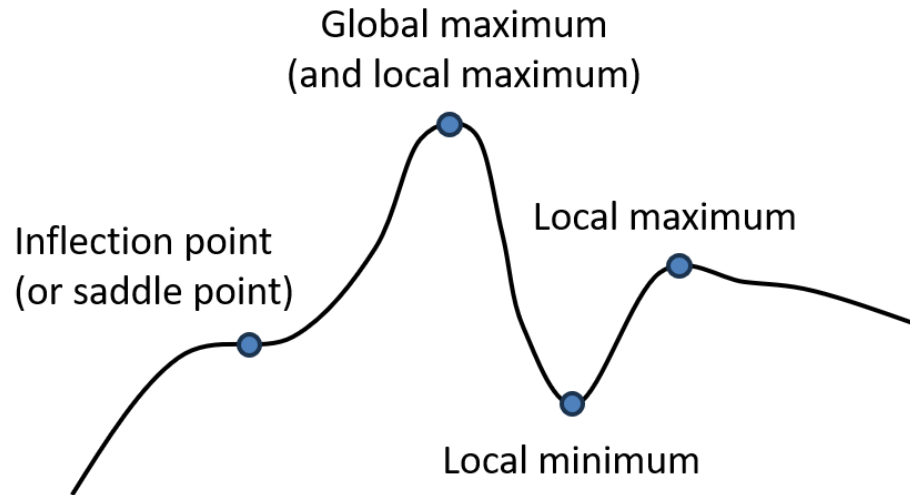
TYPE	Objective Function	Constraints	Decision Variables
Linear Program (LP)	Linear	Linear	Continuous
Nonlinear Program (NLP)	At least the objective function or 1 constraint is nonlinear		Continuous
Integer Program (IP)	Linear or nonlinear	Linear or nonlinear	All Integer
Mixed-integer Linear Program (MILP)	Linear	Linear	Some Integer; Others Continuous
Mixed-integer Nonlinear Program (MINLP)	At least the objective function or 1 constraint is nonlinear		Some Integer; Others Continuous

We'll discuss
BFGS (Unconstrained)
GA & PSO
(Unconstrained) and
IPOPT (Constrained)

Source: Chap.14; El-Halwagi, M., 2012. *Sustainable Design Through Process Integration*. USA: Elsevier Inc.

*The word "program" is due to George Dantzig (1974).

Unconstrained NLP



Local minimum

A point $\mathbf{x}^*$ such that no other point in the *vicinity* of $\mathbf{x}^*$ yields a value of $f(\mathbf{x})$ less than $f(\mathbf{x}^*)$:

$$f(\mathbf{x}) \geq f(\mathbf{x}^*)$$

Global minimum

A point $\mathbf{x}^*$ such that $f(\mathbf{x}) \geq f(\mathbf{x}^*)$ holds for any $\mathbf{x}$ in the n -dimensional space of $\mathbf{x}$.

Conditions for Optimality in Unconstrained NLPs

A point $\mathbf{x}^*$ is an **optimal solution** to an unconstrained NLP:

- ONLY IF $f(\mathbf{x})$ is twice differentiable at $\mathbf{x}^*$. **NECESSARY**
- ONLY IF $\nabla f(\mathbf{x}^*) = 0$ or $\mathbf{x}^*$ is a stationary point. **NECESSARY**
- IF $\mathbf{H}(\mathbf{x}^*)$ is *positive-definite* for a minimum to exist, and *negative-definite* for a maximum to exist. **SUFFICIENT**

$\mathbf{H}(\mathbf{x})$ is the Hessian matrix.

$$\mathbf{H}_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

If all the eigenvalues of $\mathbf{H}(\mathbf{x})$ are...

$\mathbf{H}(\mathbf{x})$ is...

Positive-definite	> 0
Positive-semidefinite	≥ 0
Negative-definite	< 0
Negative-semidefinite	≤ 0

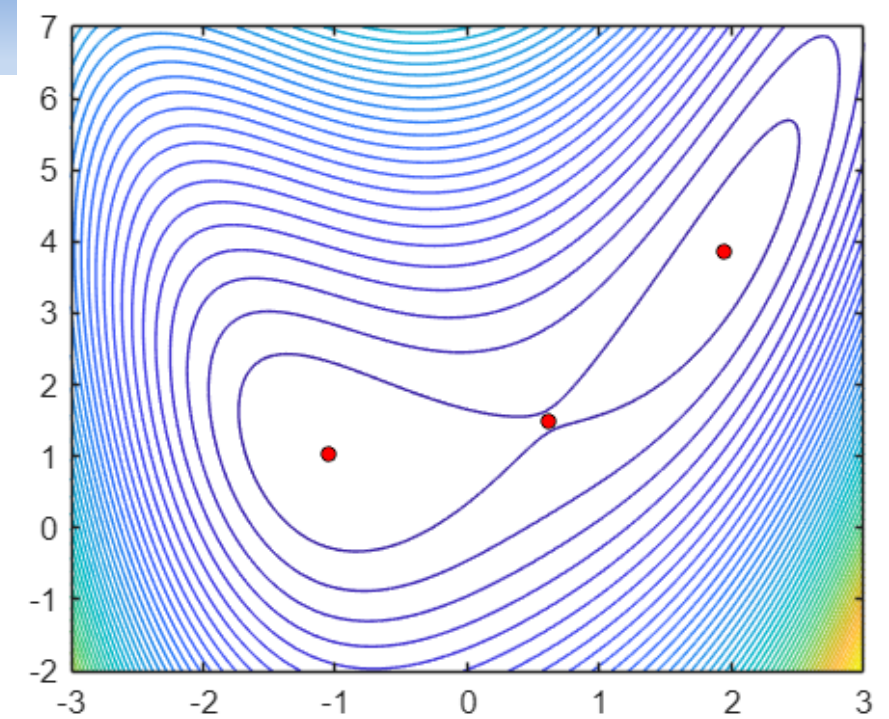
Unconstrained NLP

Example:

$$f(x_1, x_2) = 4 + 4.5x_1 - 4x_2 + x_1^2 + 2x_2^2 - 2x_1x_2 + x_1^4 - 2x_1^2x_2$$

This function has three stationary points (1.941, 3.854), (-1.053, 1.028), and (0.6117, 1.4929). Evaluate the optimality at these points.

$$H(x_1, x_2) = \begin{bmatrix} 2 + 12x_1^2 - 4x_2 & -2 - 4x_1 \\ -2 - 4x_1 & 4 \end{bmatrix}$$



Stationary point	$f(x)$	$f'(x)$	$H(x)$	Eigenvalues		Evaluation
(1.941, 3.854)	0.9856	0.00	$\begin{bmatrix} 31.79 & -9.76 \\ -9.76 & 4.00 \end{bmatrix}$	0.9128	34.8810	Local minimum
(-1.053, 1.028)	-0.5134	0.00	$\begin{bmatrix} 11.19 & 2.21 \\ 2.21 & 4.00 \end{bmatrix}$	3.3743	11.8194	Local minimum
(0.6117, 1.4929)	2.8091	0.00	$\begin{bmatrix} 0.52 & -4.45 \\ -4.45 & 4.00 \end{bmatrix}$	-2.5161	7.0346	Saddle point

Unconstrained NLP

Unlike an LP, solving an NLP is hard because a minimum is **not guaranteed** to be the global minimum.

But if the NLP is **convex**, we have the following result.

Theorem

For an NLP whose objective function $f(x)$ is a **convex function** and each inequality constraint $g(x)$ is a convex function so that they form a **convex set**:

The local minimum of $f(x)$ is also a global minimum.

Analogously, if $f(x)$ is a **concave function** and the constraints $g(x)$ still form a **convex set**, then:

The local maximum of $f(x)$ is also a global maximum.

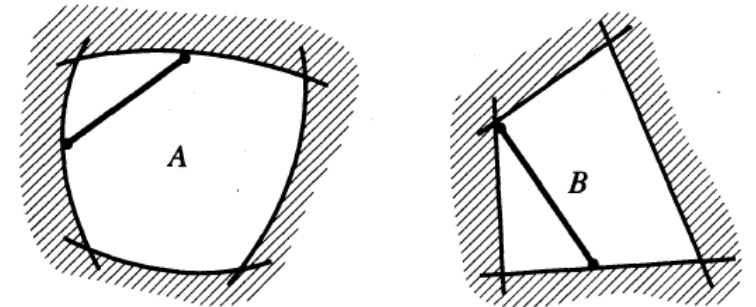
Definition: Convex Set

For every pair of points x_1 and x_2 in a convex set, the point x given by a linear combination of the two points:

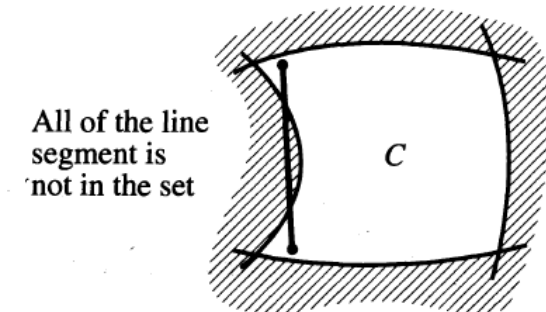
$$x = \gamma x_1 + (1 - \gamma)x_2, \quad 0 \leq \gamma \leq 1$$

must also be in the set. Also, the intersection of any number of convex sets is a convex set.

Examples of a
Convex Set



Example of a
Nonconvex Set



Unconstrained NLP

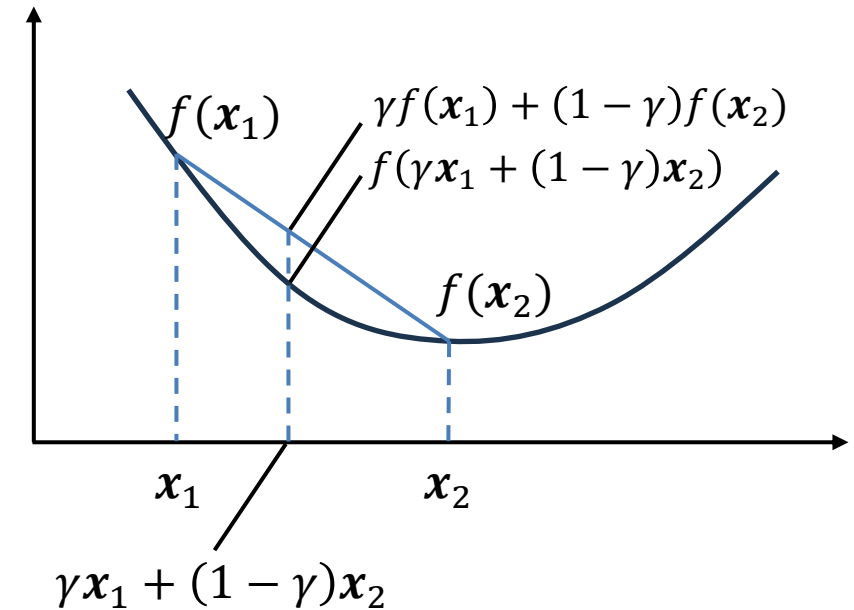
Definition: Convex Function

A function $f(x)$ defined on a convex set F is said to be a **convex function** if the following relation holds:

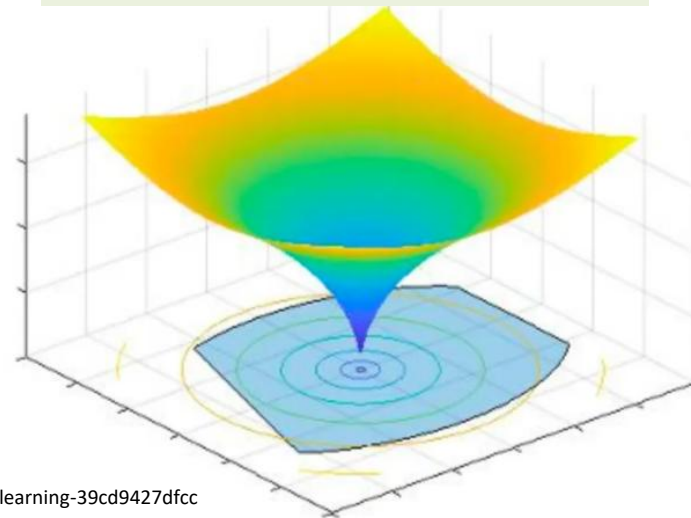
$$f(\gamma x_1 + (1 - \gamma)x_2) \leq \gamma f(x_1) + (1 - \gamma)f(x_2)$$

Where $0 \leq \gamma \leq 1$.

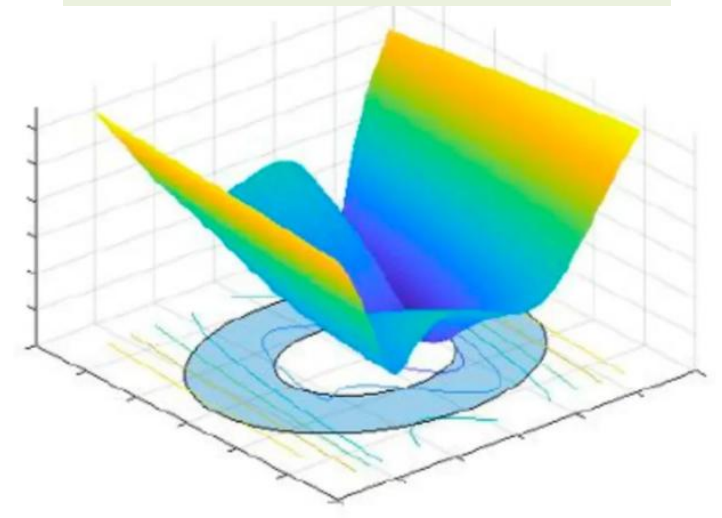
- If $f(x)$ is convex, then $-f(x)$ is concave.
- If only the inequality sign holds ($<$), the function is said to be *strictly convex*.
- If $f(x)$ is strictly convex, then $-f(x)$ is strictly concave.



A **convex** objective with
convex constraints.

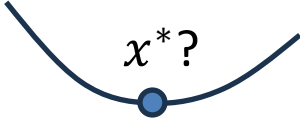


A **nonconvex** objective with
nonconvex constraints.

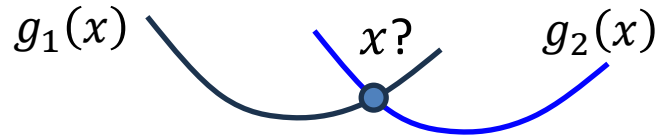


Unconstrained NLP: Newton's Methods

Minimize: $f(x)$



Recall: Newton's Method for Root-Finding



Given: $f(x) = g_1(x) - g_2(x) = 0$

Initialization: One initial guess (x_0)
and a Tolerance, Tol

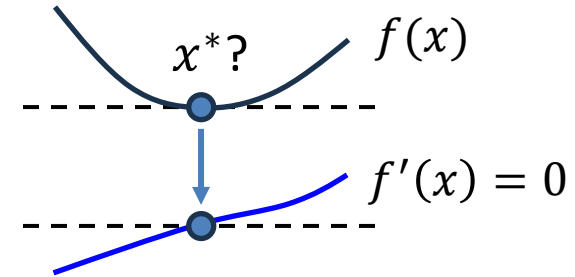
- 1: Set $x_{old} \leftarrow \text{Inf}$.
- 2: Set $x_{new} \leftarrow x_0$.
- 3: **WHILE** $|x_{new} - x_{old}| > Tol$:
- 4: Assign: $x_{old} \leftarrow x_{new}$.
- 5: Assign: $x_{new} \leftarrow x_{old} - \frac{f(x_{old})}{f'(x_{old})}$.
- 6: **END WHILE**
- 7: **OUTPUT** x_{new}

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Newton's Method for Optimization

Minimize: $f(x)$

Find the roots:
 $f'(x) = 0$



Newton's
Method

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

Quasi-Newton Method

$$x_{k+1} = x_k - \frac{(x_{k-1} - x_k)f'(x_k)}{f'(x_{k-1}) - f'(x_k)}$$

$$f''(x_k) \approx \frac{f'(x_{k-1}) - f'(x_k)}{x_{k-1} - x_k} \quad \text{2nd Order finite difference}$$

Unconstrained NLP: Newton's Methods

This formula only works for one-dimensional search spaces:

Quasi-Newton Method

$$x_{k+1} = x_k - \frac{(x_{k-1} - x_k)f'(x_k)}{f'(x_{k-1}) - f'(x_k)}$$

$$f'(x_k) \Rightarrow \nabla f(x_k)$$

$$f''(x_k) \Rightarrow H(x_k)$$

L-BFGS Algorithm

(Limited-memory Broyden-Fletcher-Goldfarb-Shanno)

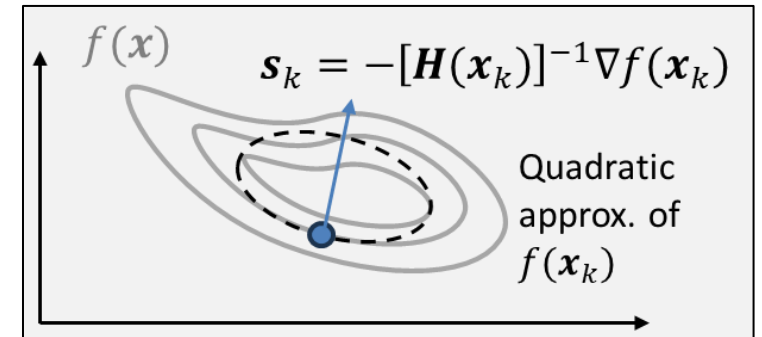
$$x_{k+1} = x_k - \alpha_k [H(x_k)]^{-1} \nabla f(x_k)$$

α_k = step size at the k th iteration

s_k = search direction

$\nabla f(x_k)$ = direction of steepest descent

$H(x_k)$ = Hessian matrix



Steps:

- Make an initial guess, x_0 , and initial guess of inverse Hessian $\Gamma_0 = I$. [*steepest descent*]
- Calculate *search direction*, $s_k = -\Gamma_k \nabla \text{Cost}(x_k)$
- Calculate step size, α , using line search.
- Update $x_{k+1} = x_k + \alpha s_k$.
- Update Γ_k to Γ_{k+1} .

In addition, BFGS only *approximates* the Hessian and computes a *step size* using line search.

Γ_k = k th approximation of the inverse Hessian

$$\Gamma_{k+1} = \left(I - \frac{s_k y_k^T}{y_k^T s_k} \right) \Gamma_k \left(I - \frac{y_k s_k^T}{y_k^T s_k} \right) + \frac{s_k s_k^T}{y_k^T s_k}$$

where $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$

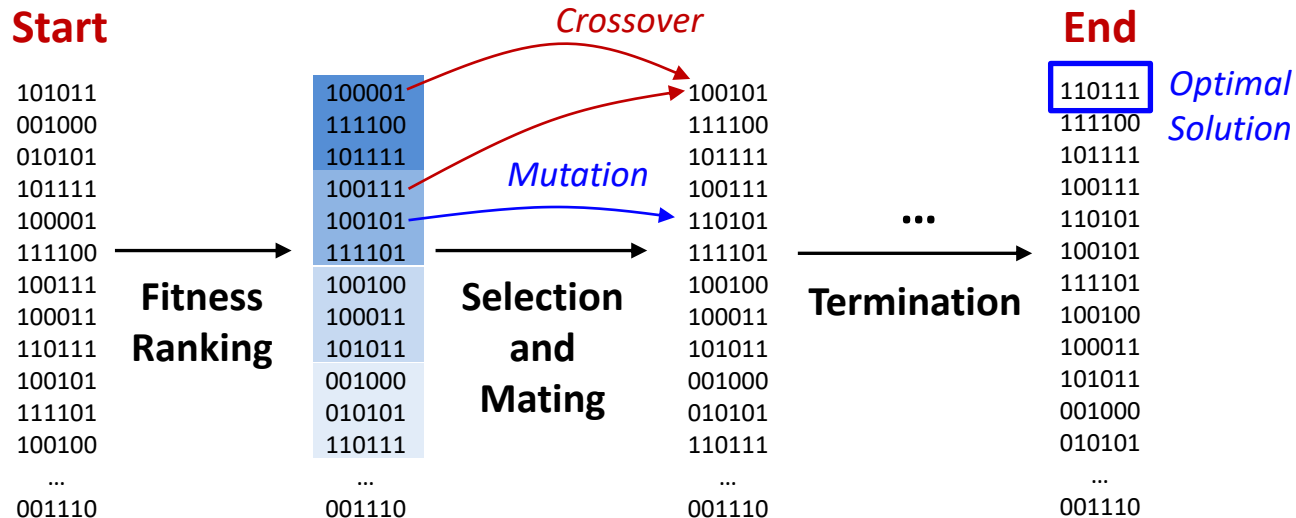
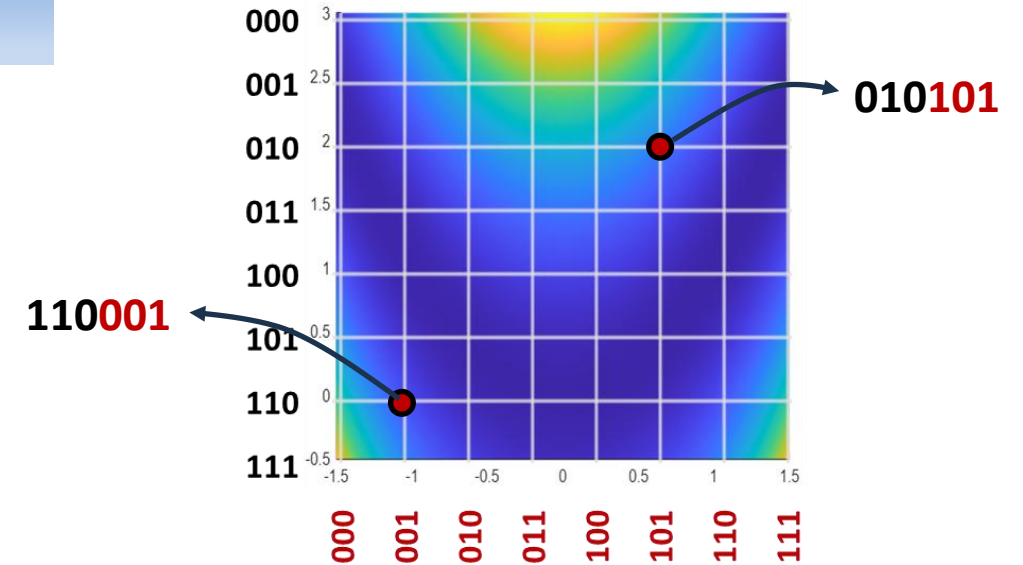
Outline

1. Motivational Example
2. Representative Algorithms
 - a. Newton Methods (Unconstrained NLP)
 - b. Evolutionary Algorithms (Unconstrained NLP)**
 - c. Lagrange Multipliers (Constrained NLP)
 - d. IPOPT (Constrained NLP)
3. Examples in Energy Engineering
 - a. Nonlinear Economic Dispatch
 - b. Economic Dispatch with 2D Valve-Point Effects
 - c. Wind Farm Layout Optimization

Evolutionary Algorithms

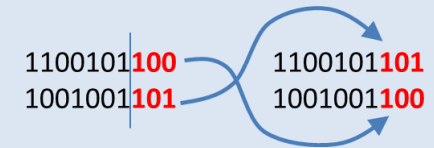
Genetic Algorithm

- John Holland, *Adaptation in Natural and Artificial Systems* (1975)
- Inspired by organisms “optimized” through years of evolution via **survival of the fittest**.
- Represent solutions as **chromosomes** (e.g. bit strings).
- In every generation, chromosomes are allowed to mate (**crossover**) and mutate (**mutation**) until convergence.



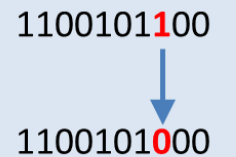
Crossover

- Promotes **exploitation**.
- Breed parents.
- The higher the fitness, the higher chance of being selected as parent.



Mutation

- Promotes **exploration**.
- Select a chromosome, then randomly change 1 or some bits.



Evolutionary Algorithms

Particle Swarm Optimization

- James Kennedy, Russell Eberhart, Yuhui Shi (1995)
- Inspired by **swarming of bees** and **flocks of birds** that can navigate together by sharing nearby information.
- Update solutions by computing their velocities, V_p^i :

$$\mathbf{x}_p^i = \mathbf{x}_p^{i-1} + \mathbf{V}_p^i$$

$$\mathbf{V}_p^i = c_0 \mathbf{V}_p^{i-1} + c_1 \times \text{rand} \times (\mathbf{x}_p^{\text{best}} - \mathbf{x}_p^{i-1}) + c_2 \times \text{rand} \times (\mathbf{x}_g^{\text{best}} - \mathbf{x}_p^{i-1})$$

c_0 Inertia weight

c_1 Cognitive parameter

c_2 Social parameter

V_p^i Velocity for the current iteration (i)

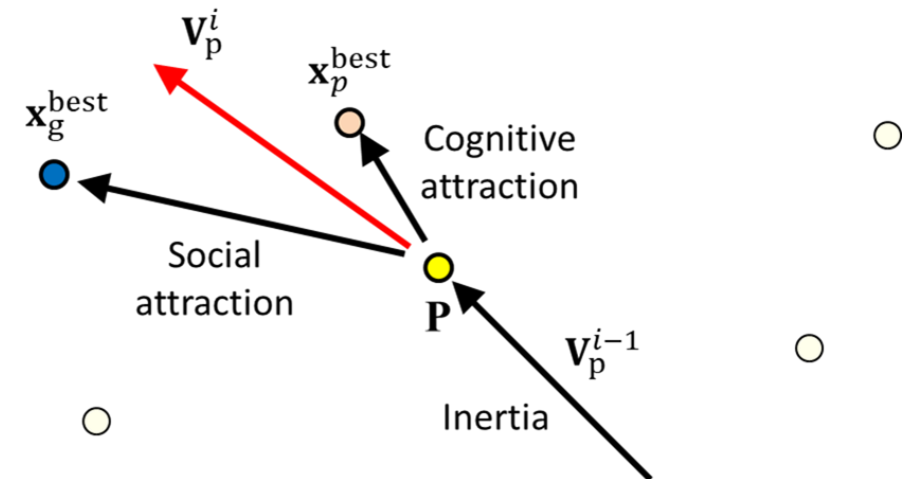
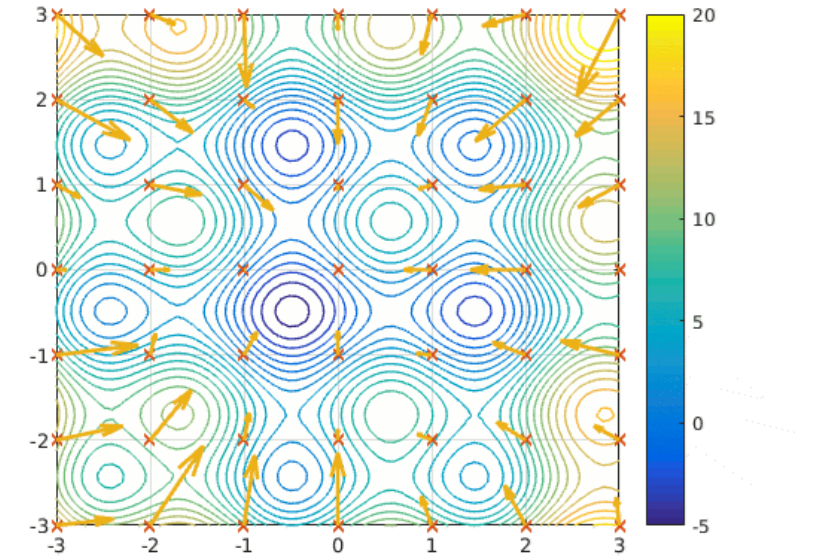
V_p^{i-1} Velocity from the previous iteration ($i - 1$)

x_p^{i-1} Previous position of the particle ($i - 1$)

x_p^{best} Best position the individual particle p encountered so far

x_g^{best} Global best position encountered by whole swarm so far

rand a random real number from 0 to 1 that's different every time.

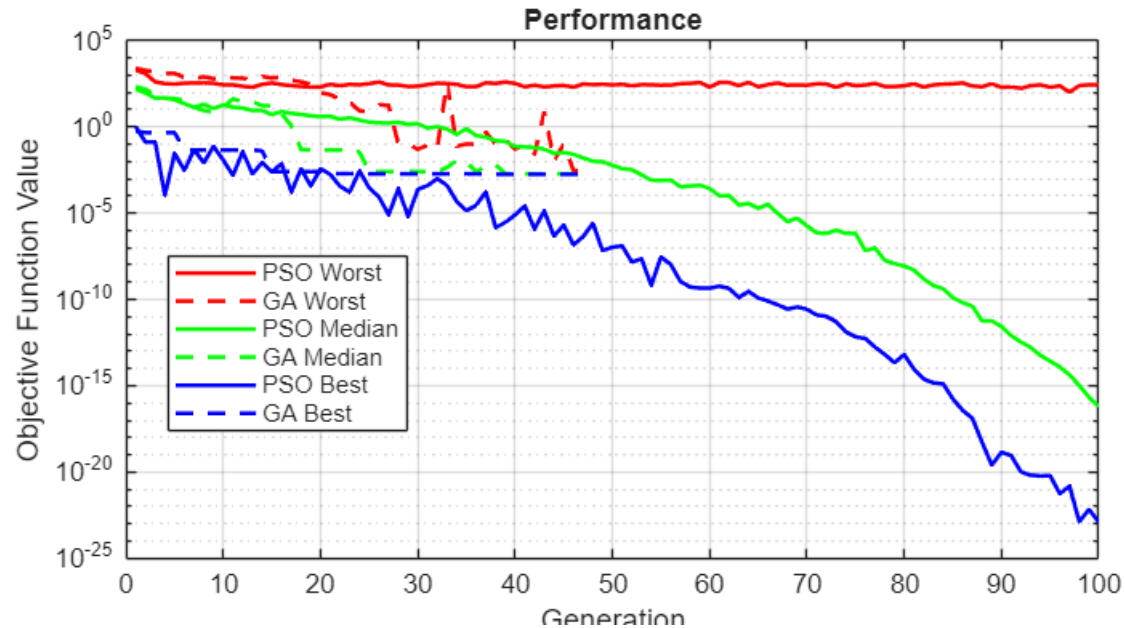


Evolutionary Algorithms

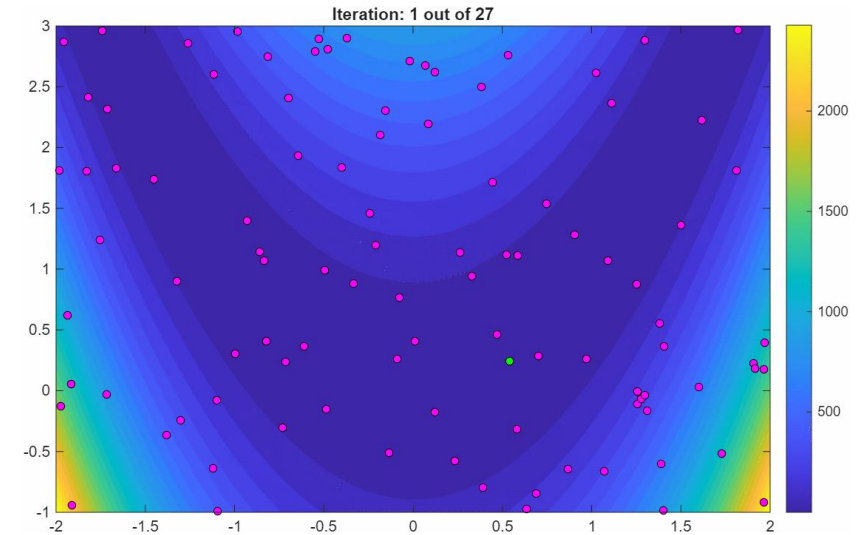
Example:

Find the minimum of the **Rosenbrock function** using GA and PSO with the following settings:

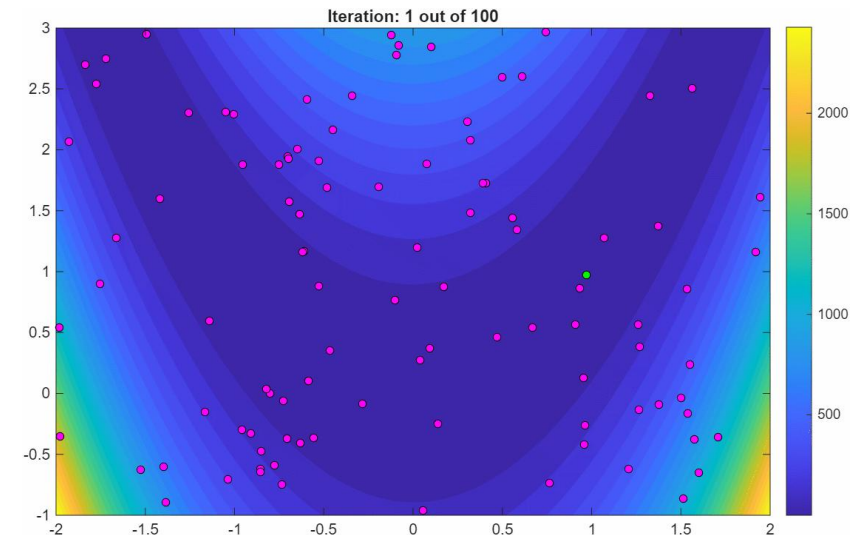
- Population size: 100 (both)
- No. of iterations: 100 (both)
- GA: Crossover rate = 1.0, Mutation rate = 0.01
- PSO: Cognitive = 2.0, Social = 2.0, inertia = 0.9 to 0.1



Genetic Algorithm



Particle Swarm Optimization



Evolutionary Algorithms

By August 2020, there are already more than 200 different nature-inspired optimization algorithms!

Source: Tzanetos et al. (Aug 2020). A comprehensive database of Nature-Inspired Algorithms. Data in Brief. Elsevier. doi: 10.1016/j.dib.2020.105792

Ant Colony Optimization	Termite Colony Optimization	Selfish Herd Optimizer	Mussels Wandering Optimization	Cheetah Chase Algorithm	Chemotherapy Science Algorithm	Artificial Physics Optimization
Bacterial Foraging Optimization	Krill Herd	Bison	Prey-Predator Algorithm	Falcon Optimization Algorithm	Kidney-inspired Algorithm	Charged System Search
BeeHive	Social Spider Optimization	Biology Migration Algorithm	Magnetotactic Bacteria Optimization Algorithm	Flying Squirrel Optimizer	Artificial Coronary Circulation System	Chemical-reaction-inspired Optimization
Bee Colony Optimization	Swallow Swarm Optimization Algorithm	Lion Pride Optimization Algorithm	Symbiotic Organisms Search	Harris Hawks Optimization	Artificial Ecosystem-based Optimization	Grenade Explosion Method
The Bees Algorithm	Spider Monkey Optimization Algorithm	Wilbeests Herd Optimization	Coral Reefs Optimization Algorithm	Sooty Tern Optimization Algorithm	Space Gravitational Algorithm	Ray Optimization
Artificial Bee Colony	Wolf Pack Algorithm	Artificial Social Cockroaches	Bird Mating Optimizer	Donkey and Smuggler Optimization Algorithm	Big Bang - Big Crunch	Mine Blast Optimization
Bee Swarm Optimization	Earthworm Optimization Algorithm	Meerkat-inspired Algorithm	Competition over Resources	Artificial Feeding Birds	Gravitational Search Algorithm	Spiral Dynamics Algorithm
Hunting Search	African Buffalo Optimization	Seagull Optimization Algorithm	Ant Lion Optimizer	Squirrel Search Algorithm	Galaxy-based Search Algorithm	Gases Brownian Motion Optimization
Bat Algorithm	Elephant Herding Optimization	Vocalization of Humpback Whale Optimization Algorithm	Artificial Algae Algorithm	Andean Condor Algorithm	Gravitational Interactions Optimization	Colliding Bodies Optimization
Cockroach Swarm Optimization	Elephant Search Algorithm	Military Dog Optimizer	Jaguar Algorithm	Genetic Algorithm	Black Hole	Kinetic Gas Molecule Optimization
Eurygaster Algorithm	See-See Partridge Chicks Optimization	Fitness Dependent Optimizer	Sperm Whale Algorithm	Bald Eagle Search	Space Gravity Optimization	States of Matter
Blind naked Mole-rat Algorithm	Dolphin Swarm Algorithm	Collective Animal Behavior	Virus Colony Search	Raccoon Optimization Algorithm	Multi-verse Optimizer	General Relativity Search Algorithm
OptBees	Lion Optimization Algorithm	Side-Blotched Lizard Algorithm	Shark Smell Optimization	Sea Lion Optimization Algorithm	Supernova Optimizer	Optics Inspired Optimization
Wolf Search	Emperor Penguin Optimizer	Marriage in Honey Bees	Crow Search Algorithm	Electric Fish Optimization	Simulated Annealing	Stochastic Fractal Search
African Wild Dog Algorithm	Coyote Optimization Algorithm	Bacterial Chemotaxis	Camel Algorithm	Urban Pigeon-Inspired Optimiser	Intelligent Water Drop	Heat Transfer Search
Seven-spot Ladybird Optimization	Anglerfish Algorithm	Honey-bees Mating Optimization	Virulence Optimization Algorithm	Xerus Optimization Algorithm	River Formation Dynamics	Ions Motion Algorithm
Penguins Search Optimization Algorithm	Dominion Algorithm	Shuffle frog-leaping Algorithm	Mosquito flying optimization	Artificial Ecosystem-based Optimization	Wind Driven Optimization	Sine Cosine Algorithm
Grey Wolf Optimizer	Emperor Penguins Colony	Cat Swarm Optimization	Red Deer Algorithm	Marine Predators Algorithm	Water Cycle Algorithm	Attraction Force Optimization
Chicken Swarm Optimization	Blue Monkey	Group Search Optimizer	Rhino Herd	Shuffled Shepherd Optimization Algorithm	Atmosphere Clouds Model Optimization	Electromagnetic Field Optimization
Cockroach Swarm Evolution	Black Widow Optimization	Monkey Search	Polar Bear Optimization Algorithm	Water Strider Algorithm	Artificial Raindrop Algorithm	Ideal Gas Molecular Movement Algorithm
Butterfly Optimization Algorithm	Black Widow Optimization Algorithm	Japanese tree frogs	Fish Electrolocation Optimization	Photosynthetic Algorithm	Hurricane-based Optimization	Thermal Exchange Optimization
Whale Optimization Algorithm	Stochastic Diffusion Search	Biogeography-based Optimization	Satin Bowerbird Optimizer	Saplings Growing up Algorithm	Lightning Search Algorithm	Vibrating Particles System Algorithm
Bird Swarm Algorithm	Particle Swarm Optimization	Bee Collecting Pollen Algorithm	Artificial Butterfly Optimization	Invasive Weed Optimization	Water Wave Optimization	Sonar Inspired Optimization
Dolphin Swarm Optimization Algorithm	Sheep Flocks Heredity Model Algorithm	Viral Systems	Beetle Antennae Search Algorithm	Plant Growth Optimization	Artificial Showering Algorithm	Spring Search Algorithm
Raven Roosting Optimisation Algorithm	Bacterial Swarming Algorithm	Monkey Algorithm	Killer Whale Algorithm	Paddy Field Algorithm	Circular Water Waves	Flow Regime Algorithm
Spotted Hyena Optimizer	Jumping Frogs Optimization	Fruit Fly Optimization	Laying Chicken Algorithm	Plant Propagation Algorithm	Vortex Search Algorithm	Coulombs Franklins Algorithm
Whale Swarm Algorithm	Fish School Search	Mosquitos Oviposition	Cyclical Parthenogenesis Algorithm	Artificial Plant Optimization Algorithm	Water Evaporation Optimization	Vapour-liquid Equilibrium-based Algorithm
Meerkat Clan Algorithm	Artificial Searching Swarm Algorithm	Bat Sonar Algorithm	Cyclical Parthenogenesis Algorithm	Ecology-inspired Optimization Algorithm	Crystal Energy Optimization Algorithm	Atom Search Optimization
Yellow Saddle Goatfish	Hierarchical Swarm Model	Hoopoe Heuristic Optimization	Beetle Swarm Optimization Algorithm	Flower Pollination Algorithm	Lightning Attachment Procedure Optimization	Henry Gas Solubility Optimization
Sailfish Optimizer	Bumble Bees Mating Optimization	Lion Pride Optimizer	Eagle Piercing Optimizer	Physarum Optimization	Rainfall Optimization Algorithm	Artificial Electric Field Algorithm
Manta Ray Foraging Optimization	Migrating Birds Optimization	The Great Salmon Run	Rhinoceros Search Algorithm	Root Mass Optimization	Hydrological Cycle Algorithm	Evolutionary Centers Algorithm
Tunicate Swarm Algorithm	Weightless Swarm Algorithm	The Lion's Algorithm	Dolphin Pod Optimization	Forest Optimization Algorithm	Nuclear Fission–Nuclear Fusion	Ohm's Law Optimization algorithm
Fish-swarm Algorithm	Pigeon-inspired Optimization	Superbug Algorithm	Owl Search Algorithm	Root Growth Algorithm	Nuclear Reaction Optimization	Electron Radar Search Algorithm
Glowworm Swarm-based Optimization	Animal Migration Optimization	Keshtel Algorithm	Pontogammarus Maeoticus Swarm Optimization	Seed-based Plant Propagation Algorithm	Whirlpool Algorithm	Spherical Search Optimizer
Roach Infestation Optimization	Moth Flame Optimization	Swine Influenza Models Based Optimization	Barnacles Mating Optimizer	Tree Seed Algorithm	Electrostatic Discharge Algorithm	Equilibrium Optimizer
Bumblebees	Cockroach Colony Optimization	Dolphin Echolocation	Mouth Brooding Fish algorithm	Runner-root Algorithm	Targeted Showering Optimization	Newton Metaheuristic Algorithm
Cuckoo Search	Monarch Butterfly Optimization	Green Herons Optimization Algorithm	Pity Beetle Algorithm	Natural Forest Regeneration	Harmony Search	Tree Growth Algorithm
Dolphin Partner Optimization	Dragonfly Algorithm	Egyptian Vulture Optimization Algorithm	Cheetah Based Optimization Algorithm	Waterweeds Algorithm	Hysteretic Optimization	Farmland Fertility Algorithm
Firefly Algorithm	Grasshopper Optimization Algorithm	Jumper Firefly Algorithm	Circular Structures of Puffer Fish	Root Tree Optimization Algorithm	Electromagnetism-like Optimization	Tree Based Optimization
Locust Swarms	Salp Swarm Algorithm	Cuttlefish Algorithm	Water-Tank Fish Algorithm	Artificial Tree Algorithm	Particle Collision Algorithm	BladderWorts Suction Algorithm
			Neuronal Communication	Mushroom Reproduction Optimization	Central Force Optimization	Physarum-energy Optimization Algorithm
				Invasive Tumor Growth Optimization	Zombie Survival Optimization	Fertile Field Algorithm
				Allostatic Optimization	Stem Cells Optimization Algorithm	Clonal Selection Algorithm
				Artificial Infectious Disease Optimization	Heart algorithm	Reincarnation Algorithm

Outline

1. Motivational Example
2. Representative Algorithms
 - a. Newton Methods (Unconstrained NLP)
 - b. Evolutionary Algorithms (Unconstrained NLP)
 - c. **Lagrange Multipliers (Constrained NLP)**
 - d. IPOPT (Constrained NLP)
3. Examples in Energy Engineering
 - a. Nonlinear Economic Dispatch
 - b. Economic Dispatch with 2D Valve-Point Effects
 - c. Wind Farm Layout Optimization

Theory of Lagrange Multipliers

Consider a general **constrained NLP**:

Minimize: $f(\mathbf{x})$

Subject to: $h_i(\mathbf{x}) = 0 \quad i \in \mathcal{E}$

$g_j(\mathbf{x}) \leq 0 \quad j \in \mathcal{J}$

and $x_k^L \leq x_k \leq x_k^U \quad k = 1, 2, \dots, n$

If $\mathbf{x}^*$ is a local minimizer and the *gradients of the active constraints* satisfy linear independence (LICQ), then there exists a set of scalars (**Lagrange multipliers**), λ_i ($i \in \mathcal{E}$) and σ_j ($j \in \mathcal{J}$) such that:

Karush-Kuhn-Tucker (KKT) Conditions

$$\nabla f(\mathbf{x}^*) + \sum_{i \in \mathcal{E}} \lambda_i \nabla h_i(\mathbf{x}^*) + \sum_{j \in \mathcal{J}} \sigma_j \nabla g_j(\mathbf{x}^*) = 0$$

$$\text{or } \nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\sigma}^*) = 0 \quad (\text{stationarity})$$

$$h_i(\mathbf{x}^*) = 0 \quad (\text{primal feasibility})$$

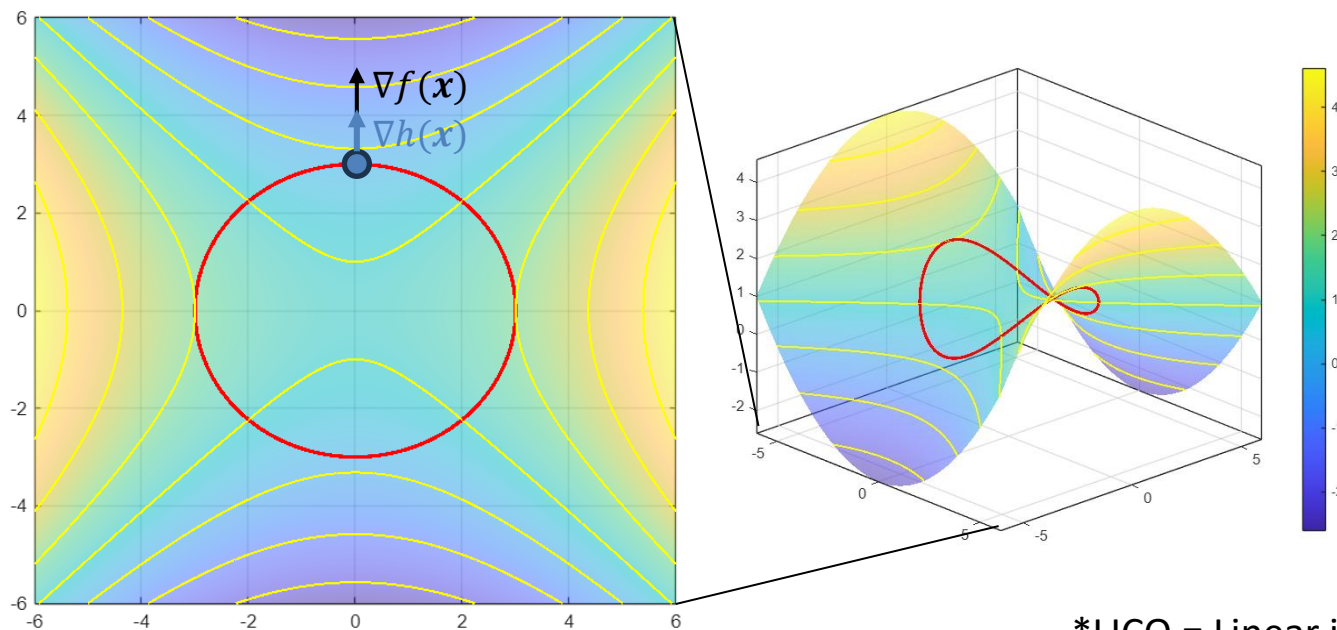
$$g_j(\mathbf{x}^*) \leq 0$$

$$\lambda_i \geq 0, \quad i \in \mathcal{E} \quad (\text{dual feasibility})$$

$$\sigma_j \geq 0, \quad j \in \mathcal{J}$$

$$\sigma_j g_j(\mathbf{x}^*) = 0, \quad j \in \mathcal{J} \quad (\text{complementarity})$$

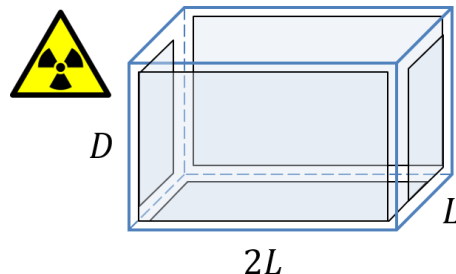
*LICQ = Linear independence constraint qualification



Theory of Lagrange Multipliers

Solve our previous example analytically using Lagrange Multipliers:

$$\min 1200(6DL + 2L^2)$$
$$2DL^2 = 8$$



Form the Lagrangian: $\mathcal{L}(D, L, \lambda) = f(D, L) - \lambda h(D, L)$

$$f(D, L) = 1200(6DL + 2L^2)$$

$$h(D, L) = 2DL^2 - 8$$

$$\mathcal{L}(D, L, \lambda) = 1200(6DL + 2L^2) - \lambda(2DL^2 - 8)$$

According to the KKT conditions:

$$\boxed{\nabla f(\mathbf{x}^*) + \sum_{i \in \mathcal{E}} \lambda_i \nabla h_i(\mathbf{x}^*)} + \sum_{j \in \mathcal{J}} \sigma_j \nabla g_j(\mathbf{x}^*) = 0$$

Solve for all gradients and equate to zero:

$$\frac{d\mathcal{L}}{dD} = 7200L - \lambda L^2 = 0$$

$$\frac{d\mathcal{L}}{dL} = 7200D + 4800L - 2\lambda DL = 0$$

$$\frac{d\mathcal{L}}{d\lambda} = -(DL^2 - 4) = 0$$

Solve for the system of 3 equations:

$L = 1.82$
$D = 1.21$
$\lambda = 3962.31$

Outline

1. Motivational Example
2. Representative Algorithms
 - a. Newton Methods (Unconstrained NLP)
 - b. Evolutionary Algorithms (Unconstrained NLP)
 - c. Lagrange Multipliers (Constrained NLP)
 - d. IPOPT (Constrained NLP)**
3. Examples in Energy Engineering
 - a. Nonlinear Economic Dispatch
 - b. Economic Dispatch with 2D Valve-Point Effects
 - c. Wind Farm Layout Optimization

A Taxonomy of NLP Solvers

Zero-order Methods

Derivative-free, black-box

- Grid Search / Exhaustive Search
- Random Search
- Nelder-Mead Simplex
- Metaheuristic Search
 - **Genetic Algorithms**
 - **Particle Swarm**
 - Simulated Annealing
 - Differential Evolution
 - CMAES
- **Bayesian Optimization / Surrogate-based**

First-order Methods

Uses 1st derivative information

- Steepest Descent or Gradient Descent
- Stochastic Gradient Descent (SGD)
 - RMSProp, Adam, etc.
- Conjugate Gradient methods (e.g. Fletcher-Reeves)
- Coordinate Descent
- Generalized Reduced Gradient (GRG)

Second-order Methods

Uses 1st and 2nd derivative information

- *Newton's method*
- **Quasi-newton method (BFGS)**
- Levenberg-Marquardt
- Trust Region Newton Methods
- Active Set Methods
- Sequential Quadratic Programming (SQP)
- **Interior Point Optimizer (IPOPT)**

Active Set vs. Interior Point Methods

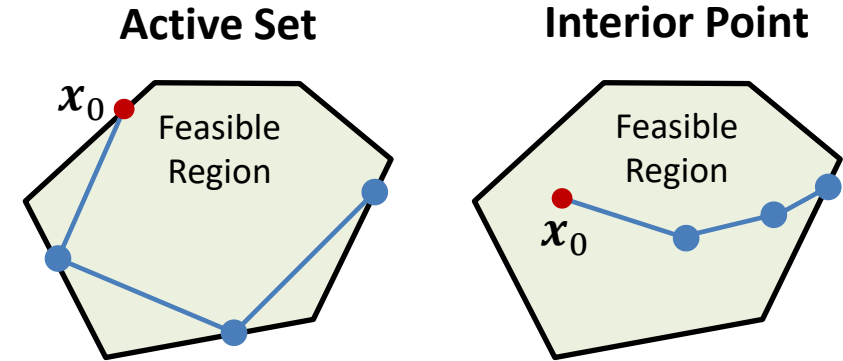
Second-order Methods

Uses 1st and 2nd derivative information

- *Newton's method*
- *Quasi-newton method (BFGS)*
- Levenberg-Marquardt
- Trust Region Newton Methods
- Active Set Methods
- Sequential Quadratic Programming (SQP)
- **Interior Point Optimizer (IPOPT)**

Active Set and Interior Point Methods

$$\begin{aligned} \text{Minimize:} \quad & f(x) \\ \text{Subject to:} \quad & h_i(x) = 0 \quad i = 1, 2, \dots, l \\ & g_j(x) \leq 0 \quad j = 1, 2, \dots, m \\ \text{and} \quad & x_k^L \leq x_k \leq x_k^U \quad k = 1, 2, \dots, n \end{aligned}$$



Active Set Methods

- Popular since the 1970s
- **Idea:** Guess an initial active set of constraints, then add / delete using gradient and Lagrange multiplier information until optimality is detected.
- **Suitable for:** small number of constraints
- **Usage:** MATLAB SQP, Python scipy SLSQP, etc.

Interior Point Methods

- Popular since the 1990s
- **Idea:** Guess an initial x inside the feasible region, then stay strictly inside as you iterate until a boundary is approached.
- **Suitable for:** Large-scale, sparse problems
- **Usage:** IPOPT, MOSEK, other modern solvers

IPOPT (Interior Point OPTimizer)

<https://github.com/coin-or/lpopt>

- Designed to find **local solutions** of optimization problems of the form:

$$\begin{aligned} & \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \\ \text{s.t. } & \mathbf{g}_L \leq \mathbf{g}(\mathbf{x}) \leq \mathbf{g}_U \\ & \mathbf{x}_L \leq \mathbf{x} \leq \mathbf{x}_u \end{aligned}$$

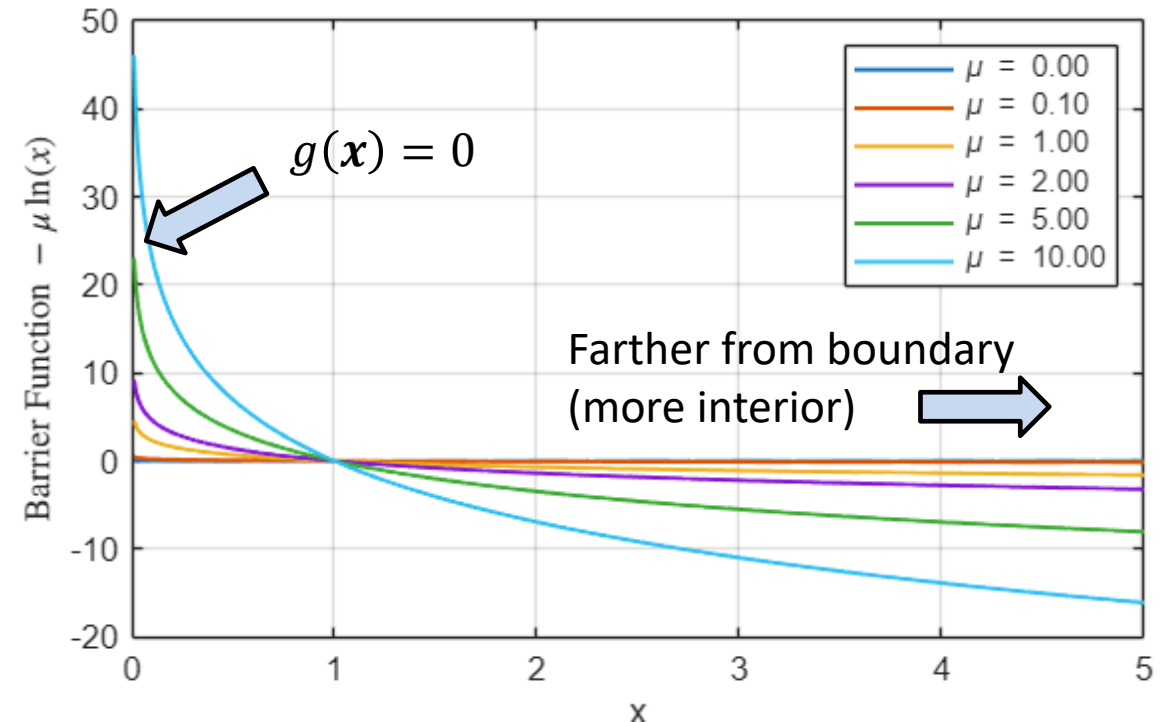
- The functions $f(\mathbf{x})$ and $\mathbf{g}(\mathbf{x})$ can be nonlinear, nonconvex, but must be twice differentiable.
- Main steps:
 - Initialization
 - Check convergence
 - Compute search direction: **Damped Newton**
 - Solve for step size: **Filter Line Search**
 - Update
- A. Wächter and L. T. Biegler, On the Implementation of a Primal-Dual Interior Point Filter Line Search Algorithm for Large-Scale Nonlinear Programming, *Mathematical Programming* 106(1), pp. 25-57, 2006.

Barrier Method

Transform the NLP by adding a log-penalty, then decrease the value of *barrier parameter*, μ , to approach the original NLP.

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \longrightarrow \quad \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) - \mu \sum_i \ln(g(\mathbf{x}_i))$$

$$\text{s.t. } g(\mathbf{x}) \geq 0$$

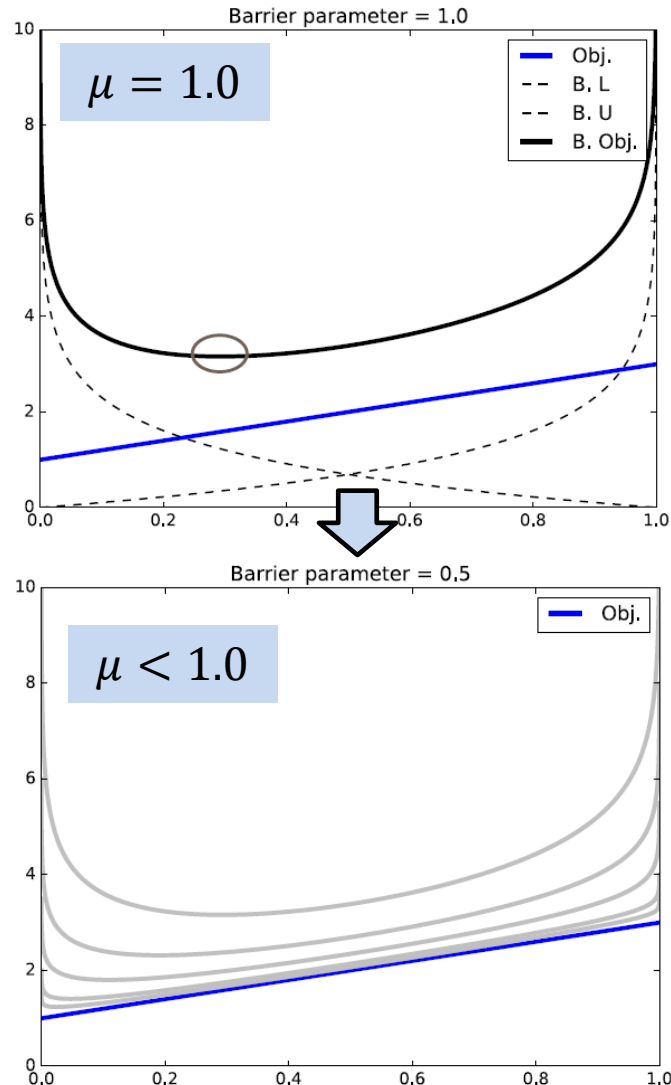


Source: <https://www.youtube.com/watch?v=bJ0Kkf4u9bo>

IPOPT (Interior Point OPTimizer)

<https://github.com/coin-or/Ipopt>

- Effect of the barrier term:

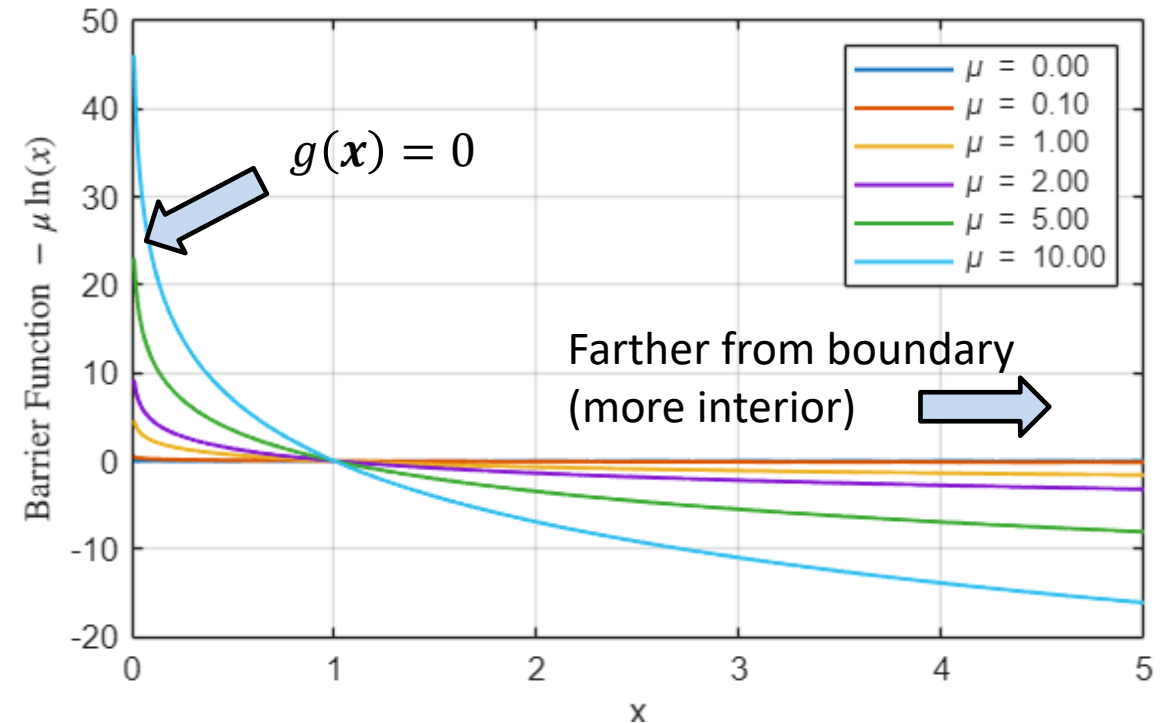


Barrier Method

Transform the NLP by adding a log-penalty, then decrease the value of *barrier parameter*, μ , to approach the original NLP.

$$\min_{x \in \mathbb{R}^n} f(x) \quad \longrightarrow \quad \min_{x \in \mathbb{R}^n} f(x) - \mu \sum_i \ln(g(x_i))$$

$$\text{s.t. } g(x) \geq 0$$



Source: <https://www.youtube.com/watch?v=bJ0Kkf4u9bo>

Outline

1. Motivational Example
2. Representative Algorithms
 - a. Newton Methods (Unconstrained NLP)
 - b. Evolutionary Algorithms (Unconstrained NLP)
 - c. Lagrange Multipliers (Constrained NLP)
 - d. IPOPT (Constrained NLP)
3. Examples in Energy Engineering
 - a. **Nonlinear Economic Dispatch**
 - b. **Economic Dispatch with 2D Valve-Point Effects**
 - c. **Wind Farm Layout Optimization**